

AUDITABLE BASIS-RELATIVE RESPONSE COORDINATES FOR FEW-SHOT SCIENTIFIC CURVES

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ABSTRACT

Few-shot scientific curve models infer entity state from sparse measurements, but raw latent coordinates are chart-dependent: an invertible reparameterization can preserve every prediction while changing the coordinates given to symbolic regression. We instead project responses on fixed physical probes onto a named basis fixed or selected on development entities and sealed before external evaluation. The resulting coordinates are canonical only within that specification; an unseen entity’s coefficients then use support targets alone. This construction is invariant to response-equivalent representations, and stable response-metric Gauss–Newton preserves independently recalibrated predictions to 3.66×10^{-10} under controlled affine chart interventions. In single-use future cohorts, compact three-coordinate equations reach physical $R^2 = 0.9888$ on 30 thermoelectric materials and $R^2 = 0.9996$ on 84 vapor-pressure compounds. Their terms describe reference level, sensitivity, and curvature or effective thermodynamic correction. Yet kNN is not significantly worse for thermoelectrics, and log-pressure PCHIP is more accurate. Our contribution is therefore a basis-relative inference protocol—not a new latent predictor or universal canonicalizer—with expression fidelity distinct from prediction and causal recovery.

1 INTRODUCTION

Many scientific measurements arrive as families of response curves rather than independent rows. A new material or compound is observed at only a few support conditions, yet the scientist wants both the remaining response curve and a compact description of how that entity differs from previously measured ones. We study this support-conditioned setting: given sparse measurements of an unseen entity, infer entity-specific coordinates from the support alone and use them to explain its hidden query response. Thermoelectric figure of merit, vapor pressure, and crystal heat capacity illustrate the central tension. A flexible predictor can interpolate these curves accurately, but a scientific coordinate must additionally be stable enough to name, compare, and place in a testable equation.

Few-shot function models and implicit neural representations already provide powerful mechanisms for predicting a new function from a context set (Garnelo et al., 2018; Xu et al., 2020; Gondal et al., 2021; Dupont et al., 2022). Their internal task vectors, however, are generally coordinates of a learned chart rather than intrinsic scientific variables. An invertible rotation, scaling, or affine change of a latent code can be absorbed by the decoder without changing a single physical prediction, while changing every individual coordinate supplied to symbolic regression. Low dimensionality alone does not ensure identifiability, and common regularized disentanglement objectives do not remove the need for explicit assumptions or inductive biases (Locatello et al., 2019; Khemahem et al., 2020). Moreover, canonicalization itself has nontrivial scope and continuity limits (Ma et al., 2024; Dym et al., 2024). Consequently, a predictive raw latent vector cannot be named as a physical quantity merely because it supports accurate reconstruction.

Scientific equation discovery faces the complementary problem. Sparse dynamics, neural-to-symbolic extraction, and parametric equation models can recover compact relations from suitable coordinates or learned components (Champion et al., 2019; Cranmer et al., 2020; Zhang et al., 2024; Podina et al., 2023). Yet when an equation coefficient is inferred for an unseen entity, the input coordinate must be determined from that entity’s support and must not depend on an arbitrary neural chart. Existing ingredients therefore leave an interface question unanswered: what representation-

level object should be interpreted when different latent parameterizations predict the same scientific response?

We answer this question with *auditable basis-relative response coordinates*. We fix a named structure a priori or select it using development entities only; when refinements are allowed, a bounded theory-guided grammar, grouped support/query scoring, and a fixed tie break select among them. Its basis and probe grid are then sealed for external evaluation. We evaluate the decoder on those probes and project its response onto that basis. Two representations that agree on the declared probe response then have exactly the same coefficients within that frozen specification, regardless of their latent dimension, redundancy, or chart. For a new entity, stable response-metric Gauss–Newton calibration uses only its support; direct structure re-estimation provides the auditable endpoint. We additionally formulate a decoder-prior diagnostic, GIRD, whose singular-direction analysis predicts benefit primarily when support is underidentified. Our controlled experiment selects its prior weight inside each predeclared support regime; it does not instantiate a per-entity rank router. This separates three questions that are often conflated: whether support contains entity state, whether that state admits named response coordinates, and whether a learned decoder prior adds information beyond the support.

The resulting interpretation target is deliberately operational. An expression succeeds when an unseen entity’s support-only coefficients yield finite physical-unit query predictions with pooled $R^2 \geq 0.85$ and its compact named terms offer a scientifically suggestive stage-wise reading. It need not recover the initial raw latent code, a preselected physical variable, or a unique microscopic law. We report coefficient stability and falsifiable follow-up tests as evidence for that reading rather than treating a particular latent chart as the target. Under a frozen temporal protocol, the quadratic response chart $ZT(T) = c_0 + c_1\tau + c_2\tau^2$ reaches $R^2 = 0.9888$ on 30 future materials, and a reference-pressure/enthalpy/curvature expression reaches $R^2 = 0.9996$ on 84 future vapor-pressure compounds. The coordinates are stable under support offsets and suggest direct tests of temperature turnover and effective vaporization-enthalpy trends. Strong local interpolation remains competitive or better—log-pressure PCHIP is more accurate for vapor pressure, and the ZT advantage over kNN is not significant—so our claim is an auditable scientific coordinate interface rather than universal predictive superiority.

Our contributions are:

- We formulate interpretation on equivalence classes of fixed physical probe responses and show that named basis coefficients are invariant to any response-equivalent representation; we separately characterize the narrower affine/full-rank conditions under which support calibration is equivariant.
- We provide a support-only, basis-relative inference protocol with explicit development-only structure selection, external freezing, rank, conditioning, gauge, leakage, coefficient-stability, and entity-tail audits, together with a rank-aware test of whether a learned decoder prior adds value beyond direct structure re-estimation.
- We evaluate compact equations under entity- and time-disjoint real-data protocols, reporting uncertainty and strong support-aware baselines separately from expression fidelity. Two domains already provide one-shot temporal confirmation; the Crystal heat-capacity stress test follows a frozen result-to-narrative rule.

2 RELATED WORK

Few-shot function representations. Conditional Neural Processes encode a context set to predict targets of a new function, while MetaFun performs iterative functional updates and function contrastive learning optimizes transferable function-level representations (Garnelo et al., 2018; Xu et al., 2020; Gondal et al., 2021). MetaSDF and Functa learn shared priors over implicit functions or compact modulations and adapt instance-specific representations from observations; Functa also demonstrates MAP completion from partial observations (Sitzmann et al., 2020; Dupont et al., 2022). These methods establish that support-conditioned function prediction, low-dimensional task variables, and test-time adaptation are not new. Our question begins where their predictive objective ends: if a task representation will enter a scientific equation, which response-derived coordinate is

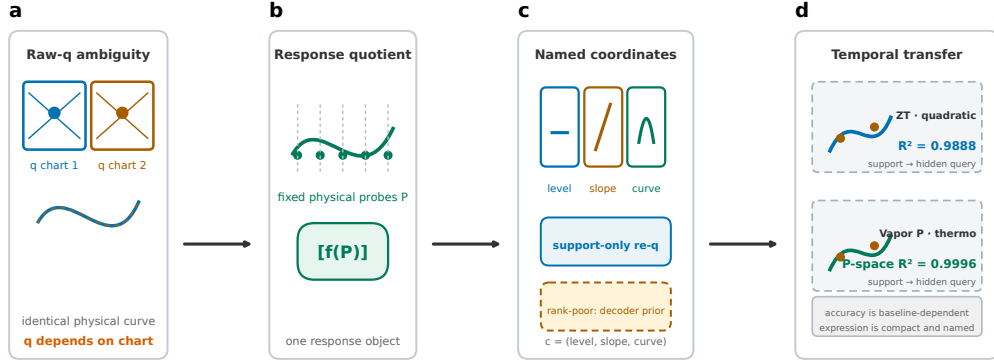


Figure 1: Raw latent coordinates can change under prediction-equivalent neural charts even when their physical curves agree (a). We therefore identify an entity through its response on fixed physical probes (b) and project that response-equivalence class onto a named basis (c). Support-only inference yields compact expressions that transfer to future thermoelectric and vapor-pressure entities (d); predictive superiority over local interpolation is evaluated separately.

invariant enough to name? We therefore evaluate CNP and functional baselines directly and interpret coefficients of a declared physical response basis rather than an unconstrained context vector.

Latent identifiability and invariant structure. Unsupervised disentanglement is generally non-identifiable without inductive bias, although conditional nonlinear-ICA models identify latent sources up to specified equivalence classes under explicit auxiliary-variable and generative assumptions (Locatello et al., 2019; Khemakhem et al., 2020). Recent work shifts attention from individual latent variables to invariant metric structures, and canonicalization provides a broader framework for constructing invariant or equivariant representations (Syrota et al., 2025; Ma et al., 2024). Canonicalization can itself be discontinuous for common group actions, motivating weighted frames and careful scope statements (Dym et al., 2024). We neither claim generic canonicalization nor recovery of true sources. Our narrower object is the equivalence class induced by agreement on declared physical probes; its fixed linear projection is continuous in the probe response, whereas the independent support optimizer retains explicit affine, rank, and conditioning assumptions.

Learned coordinates and symbolic equations. SINDy autoencoders jointly learn nonlinear coordinates and sparse governing dynamics, and neural-to-symbolic extraction recovers compact relations from structured learned components (Champion et al., 2019; Cranmer et al., 2020). Parametric equation models share symbolic structure while coefficients vary across parameterized instances, and UPINNs convert neural representations of unknown differential-equation terms into symbolic expressions from sparse observations (Zhang et al., 2024; Podina et al., 2023). LaSR uses an LLM-induced library of abstract textual concepts to guide evolutionary symbolic regression and improve benchmark performance (Grayeli et al., 2024). Accordingly, neither symbolic regression, fixed basis projection, nor shared equations with varying coefficients is our novelty. Our distinction is the audited interface around them: exact prediction-preserving chart intervention, support-only coefficients for unseen entities, response-space rank diagnostics, and single-use temporal expression evaluation with interpolation baselines and entity tails.

Together, these lines imply a deliberately bounded contribution. We combine a support-conditioned learned response, a basis-relative invariant explanatory object, and sealed symbolic transfer; we do not assert that latent variables are always unidentifiable, that our basis is a unique natural coordinate system, or that a neural prior must improve prediction. On the Crystal development registry the frozen GIRD incremental gates do not pass, so we retain it as a controlled diagnostic of when support and decoder information disagree rather than presenting it as an improved predictor.

3 FROZEN BASIS-RELATIVE RESPONSE COORDINATES

Our method maps sparse observations of an unseen entity to named equation coordinates without assigning scientific meaning to a raw neural latent chart. Entity e has observations (x, y) partitioned into support S_e and hidden query Q_e . A decoder $f_\theta(x, q_e)$ represents a family of response functions, and a new entity’s q_e is calibrated using S_e while θ is frozen. The core branch estimates equation coefficients directly from support; the optional GIRD branch asks whether the calibrated decoder supplies useful function-space information when the support design is underidentified. Both branches output the same named coefficient vector and symbolic response.

Discover then freeze. Each domain contract distinguishes choices fixed a priori from any atom selected on development entities. Only when a refinement set is allowed do grouped support/query folds score its finite, theory-guided grammar using a frozen metric and tie break. The resulting basis and all fixed scaling, probes, support rule, and gates are then sealed; this does not imply they were jointly searched. External entities receive no structure search—only support-based coefficient estimation. Thus “predeclared” means predeclared for external evaluation.

3.1 RESPONSE EQUIVALENCE AND CANONICAL COORDINATES

A raw latent coordinate is chart-dependent even when its decoded response is fixed. We therefore declare, before evaluating query targets, physical probes $P = \{x_j\}_{j=1}^m$, nonnegative weights $W = \text{diag}(w_1, \dots, w_m)$, and named basis functions $\phi(x) = [\phi_1(x), \dots, \phi_K(x)]$. Let $\Phi_P \in \mathbb{R}^{m \times K}$ contain the basis evaluated on the probes and let

$$r_\theta(q) = [f_\theta(x_1, q), \dots, f_\theta(x_m, q)]^\top \quad (1)$$

be the physical probe response. When $W^{1/2}\Phi_P$ has full column rank, we define the canonical response coordinate, canonical only within this frozen specification rather than a unique natural coordinate,

$$c_\theta(q) = (\Phi_P^\top W \Phi_P)^{-1} \Phi_P^\top W r_\theta(q). \quad (2)$$

The basis, probes, weights, units, and rank criterion are part of the method specification; changing them changes the coordinate semantics.

Proposition 1 (response-quotient invariance). Equation (2) is invariant on probe-response equivalence classes. Formally, define $(f, q) \sim_P (\tilde{f}, \tilde{q})$ when $r_f(q) = r_{\tilde{f}}(\tilde{q})$ on the declared probes. Both representations then map to the same c because the projection operator is fixed. Hence the map factors through the quotient of model-state pairs by $\sim_P$. This covers different latent dimensions, redundant states, architectures, and arbitrary invertible chart changes whenever their probe responses agree; it does not imply equality outside P , a unique natural basis, or a causal physical variable. Approximate response agreement gives the stability bound

$$\|c(r) - c(\tilde{r})\|_2 \leq \frac{\|W^{1/2}(r - \tilde{r})\|_2}{\sigma_{\min}(W^{1/2}\Phi_P)}. \quad (3)$$

Thus projection residual and basis conditioning must be audited separately from query prediction. We display the controlled cross-chart response discrepancy and retain projection residuals, the real-domain protocol, and conditioning diagnostics in the frozen artifacts.

3.2 SUPPORT-ONLY INFERENCE AND NUMERICAL EQUIVARIANCE

The auditable core estimator skips raw q and fits the named structure using only support:

$$\hat{c}_0 = \arg \min_c \|W_S^{1/2}(y_S - \Phi_S c)\|_2^2. \quad (4)$$

For $y_S = \Phi_S c^* + \epsilon_S$ and full-column-rank Φ_S ,

$$\|\hat{c}_0 - c^*\|_2 \leq \frac{\|W_S^{1/2}\epsilon_S\|_2}{\sigma_{\min}(W_S^{1/2}\Phi_S)}. \quad (5)$$

The induced noiseless-query reconstruction error obeys

$$\|\Phi_Q(\hat{c}_0 - c^*)\|_2 \leq \frac{\|\Phi_Q\|_2 \|W_S^{1/2} \epsilon_S\|_2}{\sigma_{\min}(W_S^{1/2} \Phi_S)}. \quad (6)$$

This makes the minimum singular value, condition number, query amplification, support-offset coefficient stability, and prediction tails mandatory diagnostics. High pooled R^2 alone does not identify an individual coefficient when support basis columns are nearly collinear.

To test whether a learned decoder encodes the same response coordinates, we calibrate q on S_e with response-metric Gauss–Newton. At iterate q_k , we solve $J_k \delta_k = -r_k$ by float64 direct least squares and accept a step only when it clears a response-loss decrease margin. Exact equivariance under $q' = Aq + b$ requires consistently transformed latent paths and initial centroids, a differentiable decoder with full-rank support Jacobian, and identical response loss and coordinate-free line search. Then $q'_k = Aq_k + b$ and the response paths agree. We log conditioning, steps, and paired discrepancies; the guarantee excludes nonlinear charts, rank deficiency, coordinate-norm stopping, unsynchronized paths, and ordinary Adam.

4 RANK-AWARE DECODER-PRIOR INFERENCE

4.1 GAUGE-INVARIANT RESPONSE DICTIONARY DISCOVERY

Direct support fitting is preferable when Φ_S identifies the named structure, but it can have high variance or a null space under very sparse support. GIRD tests whether a calibrated decoder response provides incremental information specifically in this regime. It never shrinks coefficients in arbitrary raw-coordinate units: all fusion occurs in the function metric defined by the fixed probes.

During development, entity/DOI-cross-fitted decoder probes drive deterministic multi-response OMP. Nested folds select dictionary width and prior weight with frozen ties; the fixed-width path is then refit on all development entities to determine atom identities without changing complexity or thresholds. In the controlled study, selection is per family–seed–support regime; rank diagnoses the regime but does not route individual test entities.

Hard atom selection needs a stronger audit than smooth projection. At each OMP step we retain the winner–runner-up margin, residual norm, and largest aligned response displacement across seeds and support offsets. Appendix A gives the preregistered sufficient margin certificate. Failure alone does not prove instability, so we combine it with atom recurrence and gauge interventions.

For an unseen entity, decoder calibration and projection yield a functional prior c_f . Define the probe Gram matrix $G = \Phi_P^\top W \Phi_P$, whitened coordinates $u = G^{1/2} c$, and

$$D = \frac{1}{\sqrt{|S|}} \Phi_S G^{-1/2}, \quad z = \frac{1}{\sqrt{|S|}} y_S. \quad (7)$$

For a frozen λ , GIRD solves

$$\begin{aligned} \hat{u}_\lambda &= \arg \min_u \{ \|Du - z\|_2^2 + \lambda \|u - u_f\|_2^2 \}, \\ \hat{c}_\lambda &= G^{-1/2} \hat{u}_\lambda. \end{aligned} \quad (8)$$

The final prediction is $\hat{y}(x) = \sum_{j=1}^K \hat{c}_{\lambda,j} \phi_j(x)$. The endpoints $\lambda = 0$ and $\lambda = \infty$ expose direct support fitting and pure decoder projection rather than hiding either source.

The singular directions of D make GIRD’s intended regime explicit. If $D = U \text{diag}(s_i) V^\top$, support noise and prior error are independent and zero mean, with directional variances σ^2 and τ_i^2 , then

$$\mathbb{E}(\hat{u}_{\lambda,i} - u_i^*)^2 = \frac{s_i^2 \sigma^2 + \lambda^2 \tau_i^2}{(s_i^2 + \lambda)^2}. \quad (9)$$

A finite prior is most useful when s_i is small and the decoder prior remains stable; it can increase error when support is already informative or the prior is biased. When $s_i = 0$, any positive λ selects the prior’s null-space coordinate rather than identifying it from support. This is a diagnostic bias–variance calculation, not a guarantee of real-data improvement.

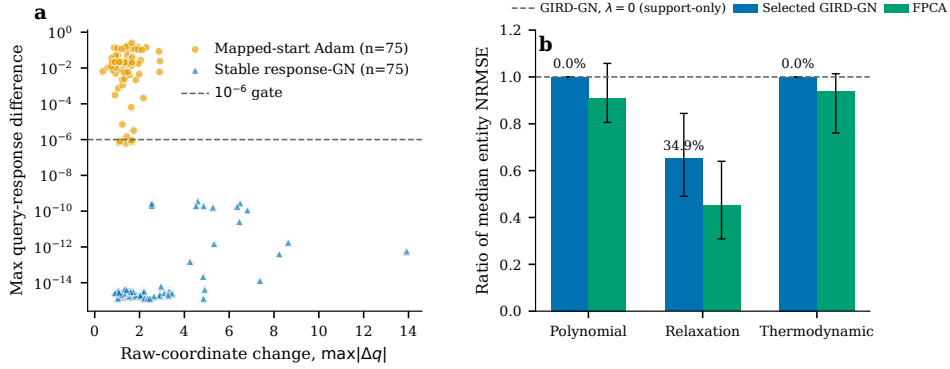


Figure 2: Stable response-metric optimization preserves gauge-equivalent predictions, while GIRD gains are family-dependent in these controlled settings. **(a)** Across 75 family–seed–gauge cases per method (3 families $\times$ 5 seeds $\times$ 5 gauges), raw coordinates move, but stable response-metric Gauss–Newton keeps the maximum query-response difference below 3.66×10^{-10} (dashed pre-specified 10^{-6} numerical tolerance), whereas mapped-start Adam reaches 0.255. **(b)** With four support points, GIRD-GN reduces median entity NRMSE only for relaxation ($0.0528 \rightarrow 0.0344$, 34.9%) and matches its $\lambda = 0$ result for polynomial and thermodynamic charts. FPCA has lower median NRMSE in all three tested families. Bars are ratios of each method’s median entity NRMSE to the support-only GIRD-GN $\lambda = 0$ median; lower is better, and error bars are paired entity-bootstrap 95% intervals (10,000 resamples) after five-seed pointwise aggregation. Hyperparameters use fixed outer-training validation; outer-test queries are untouched.

The tested GIRD variant applies validation-selected λ uniformly within each support regime; $\lambda = 0$ remains explicit, and there is no per-entity router. All choices exclude outer-test queries. The Crystal development registry compares GIRD, $\lambda = 0$, direct-target dictionary, FPCA, CNP, local interpolation, and no-latent prediction on identical partitions.

5 EXPERIMENTS

5.1 QUESTIONS AND PROTOCOL

The experiments ask four questions: (Q1) do canonical coordinates remain stable when raw latent charts change; (Q2) do compact equations transfer to unseen future entities; (Q3) does a decoder prior add value specifically when support does not identify the symbolic structure; and (Q4) what predictive accuracy is traded for a compact named expression? We split by entity and, where available, DOI and time. A test entity’s coefficients and calibrated state use support targets only; query targets are read once for scoring. Primary reporting uses physical-unit pooled R^2 , entity- or DOI-cluster bootstrap intervals, the fraction of entities above $R^2 = 0.85$, and paired comparisons. The heterogeneous Crystal stress test additionally reports median/p95/maximum entity NRMSE and negative-prediction count; every displayed aggregate requires finite full query coverage. For fitted methods with an explicit perturbation audit, perturbing hidden query targets must leave all fitted inputs, coefficients, and predictions unchanged; all such declared differences are zero. For entity e , NRMSE is query RMSE divided by the population standard deviation of its query targets (ddof = 0). Controlled five-seed GIRD predictions are aggregated pointwise before entity metrics; neural-ablation seed rules are reported with their respective results. The real-data cohorts come from the Starrydata thermoelectric collection (Katsura et al., 2025) and the NIST ThermoML archive (Riccardi et al., 2021; Frenkel et al., 2006). Frozen cohort manifests bind the selected source files by repository-local SHA-256; mutable “latest” links are not treated as the experimental artifact.

Every method is labeled by its information path. Support structure re-q fits a frozen named basis; raw-q ridge maps neural coordinates directly to equation coefficients; decoder-functional projection probes the calibrated decoder; GIRD fuses that projection using a validation-selected weight fixed within each predeclared support regime; rank and conditioning diagnose the regime rather than route individual entities. Comparators include the same-budget direct-target dictionary, FPCA (Ramsay &

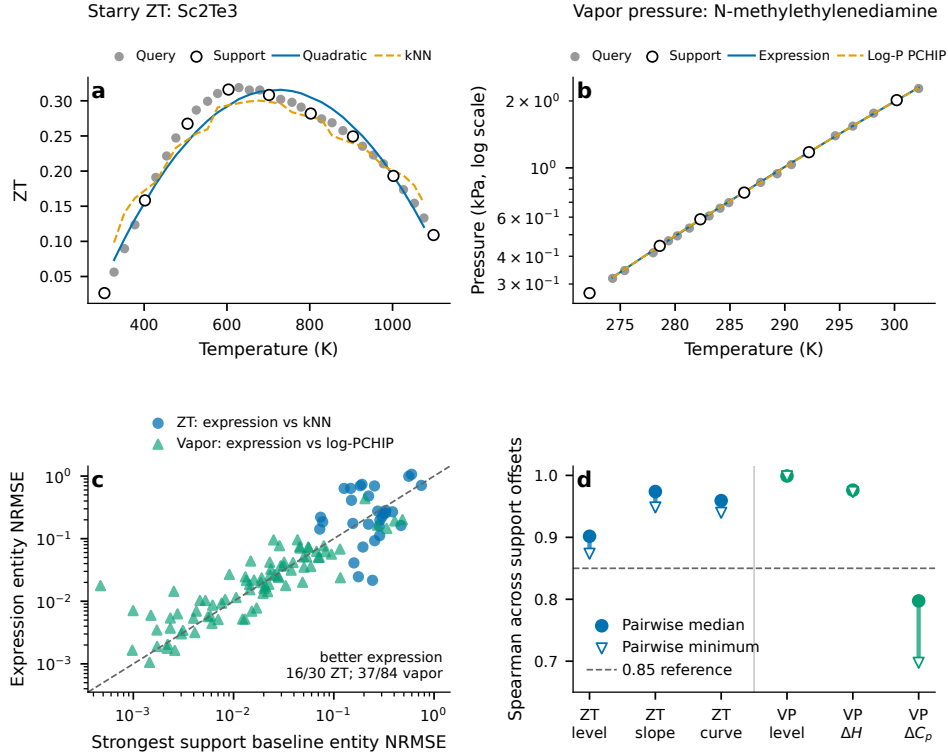


Figure 3: Compact named expressions transfer to unseen future entities, but are not uniformly more accurate than local support predictors. **(a,b)** Representative entities are selected mechanically as the entity nearest each expression family’s median NRMSE, rather than as best-case examples. Black open markers are the only targets available for coefficient fitting; gray query targets remain hidden until scoring. **(c)** Paired entity NRMSE against the strongest support-aware baseline; points below the diagonal favor the expression. The expression wins 16/30 ZT comparisons against kNN and 37/84 vapor-pressure comparisons against PCHIP, so pooled fidelity is not predictive superiority. **(d)** Median (circle) and minimum (triangle) pairwise Spearman correlations of named development coefficients over all six pairs of four support offsets; the least-stable vapor correction is interpreted only as an effective stage-wise term.

Silverman, 2005), CNP, a temperature-only no-q MLP, and support nearest-neighbor, linear, kNN, and PCHIP predictors (Fritsch & Butland, 1984). The strongest support-aware baseline is displayed beside each headline expression score. Expression fidelity and predictive superiority are tested separately.

5.2 GAUGE INTERVENTION AND CONTROLLED PRIOR EVIDENCE

We first intervene on the latent chart while preserving the decoder response. Across three controlled response families, five training seeds, and 25 exact affine gauges per family–seed cell (375 interventions), the constructed counterfactual changes raw coordinates by as much as 4.996 while changing decoder predictions and fixed-basis coefficients by at most 4.50×10^{-15} and 2.04×10^{-14} . This tests the algebraic response-quotient claim. A separate 75-case experiment independently recalibrates each of five affine charts per family–seed cell. The original response-metric Gauss–Newton extension diverges by as much as 0.0448 in query response; float64 direct least squares, a response-loss acceptance margin, and 15 fixed Gauss–Newton steps reduce the maximum response and coefficient differences to 3.66×10^{-10} and 6.23×10^{-9} , with identical paired line-search decisions. This narrower result supports numerical affine equivariance only under the stated full-rank calibration assumptions.

Table 1: One-shot unseen-entity temporal expression results. Confidence intervals resample entities, with an additional DOI-cluster interval for vapor pressure. “Pass” counts entities with individual physical $R^2 \geq 0.85$. The strongest support-aware baseline remains adjacent; higher baseline accuracy does not invalidate the separate expression endpoint. Here $\tau = (T - 280.584300 \text{ K}) / (193.386207 \text{ K})$, $T_r = 362.72 \text{ K}$, and $\Delta T^{-1} = 1/T - 1/T_r$.

Domain	Named structure	N_e	N_q	Physical $R^2 \uparrow$	$\geq .85$	Best support $R^2 \uparrow$
Starry ZT	$c_0 + c_1\tau + c_2\tau^2$	30	919	0.988810	20/30	kNN 0.982678
Vapor pressure	$c_0 + c_1\Delta T^{-1} + c_2 \log(T/T_r)$	84	1,745	0.999581	83/84	log-PCHIP 0.999937

The controlled GIRD study next varies support identifiability. Under four-point support, aggregate polynomial and thermodynamic selections match their $\lambda = 0$ results. For the relaxation family under four-point support, a finite prior reduces error from 0.05277 to 0.03435, a 34.9% development improvement. FPCA remains better than every GIRD variant, and two of 30 family-seed-regime OMP certificate audits fail. These results are consistent with the rank-conditioned bias-variance mechanism and expose its failure modes; they do not establish a universal prior advantage.

5.3 ONE-SHOT REAL-DATA EXPRESSION TRANSFER

Table 1 reports two single-use temporal confirmations. ZT’s quadratic endpoint was fixed a priori. Vapor alone used structure selection: five DOI-grouped folds scored exactly $\{\log(T/T_r), (T - T_r)/T_r, (1/T - 1/T_r)^2\}$ as one-term corrections to coarse Clausius–Clapeyron by median physical NRMSE over all 282 held-out entities; candidates within 1% use the displayed order. ZT development queries evaluate its fixed endpoint and vapor development queries select the correction. After sealing, future targets only score the structure and support estimates its coefficients (full audit in Appendix A).

Here c_i denotes a canonical response coefficient, never a raw neural latent coordinate. For ZT, c_0 , c_1 , and c_2 encode a reference value, first-order temperature sensitivity, and curvature. Entity-bootstrap physical R^2 is $[0.973306, 0.994708]$, and median entity R^2 is 0.941739. On development entities, changing the stratified-support offset gives response-coordinate distance Spearman 0.9780, 0.9733, and 0.9679 relative to offset zero; the physical curvature rank has median/minimum pairwise Spearman 0.9593/0.9397. The expression has lower NRMSE than kNN on 16/30 future materials, but the paired Wilcoxon test is nonsignificant ($p = 0.465$). Its scientific test is therefore the predicted slope reversal near $\tau^* = -c_1/(2c_2)$, not a claim of universal predictive gain.

For vapor pressure, the coordinates give reference log pressure c_0 (equivalently $P_r = (1 \text{ kPa})e^{c_0}$), effective reference vaporization enthalpy $R(c_2T_r - c_1)$, and an effective temperature correction Rc_2 . This reading uses an ideal-gas, negligible-liquid-volume, finite-temperature-range linearization and therefore defines effective response coordinates, not exact calorimetric quantities. Entity- and DOI-bootstrap physical R^2 intervals are $[0.998880, 0.999983]$ and $[0.998562, 0.999984]$. Across four development support offsets, whose fits use outer-training-fold median reference temperatures rather than the temporal confirmation’s frozen T_r , the three named-coordinate rank correlations have medians 0.9991, 0.9762, and 0.7974; joint response-coordinate geometry has median/minimum correlation 0.8905/0.8546. The weakest correction term is limited by design collinearity (each support-design column is divided by its ℓ_2 norm before the condition number; median/p95/maximum 135/917/6417) and is not presented as a calorimetric measurement. Log-pressure PCHIP has higher pooled R^2 , while the compact expression provides a global named response chart and a falsifiable comparison to independent enthalpy data.

5.4 WHAT THE NEURAL REPRESENTATION CONTRIBUTES

The neural ablations separate coordinate readability from complete bridge success. On the external vapor-pressure cohort, decoder-functional projection reaches physical $R^2 = 0.990390$, whereas a direct raw-q ridge expression reaches only 0.304492; both are pointwise medians of three seed predictions in physical pressure space. The functional projection nevertheless fails its stronger cross-seed geometry gate and is not claim-eligible as the confirmed expression; the 0.999581 result comes from support structure re-q. The same qualitative separation occurs for ZT, where each outer-fold

Table 2: Crystal-Cp development pooled physical R^2 with identical unseen- entity spread support. CNP/no-q MLP are pointwise five-seed medians; other rows are deterministic.

Method	Physical $R^2 \uparrow$
PCHIP	0.958807
Linear interpolation	0.952084
Nearest support	0.919874
Support kNN	0.898820
Boundary-stage expression	0.859300
Shomate5 support re-q	0.638646
FPCA	0.632189
CNP	0.346495
No-q temperature MLP	0.095713

prediction is the pointwise median over three seeds, fixed-basis decoder-functional fidelity is high but extreme entity-NRMSE tails prevent a universal neural-bridge claim. These ablations show that response projection repairs readability of a learned function, not that every trained decoder yields a safe equation prior.

5.5 CRYSTAL HEAT CAPACITY STRESS TEST

Crystal heat capacity tests a heterogeneous two-stage response. Ordinary curves use NIST’s five-term Shomate form (Linstrom & Mallard, n.d.); support-routed boundary curves use the nested family $A + B_1u + B_2u^2 + H/\sqrt{\delta + 1 - u}$, allowing the degree-one subcase $B_2 = 0$, where $u = (T - T_{\min,e})/(T_{\max,e} - T_{\min,e})$. Routing uses only the spread-support slope ratio (Appendix A); the sealed confirmation package freezes $(\gamma, d, \delta) = (100, 2, 3 \times 10^{-4})$. Coefficients denote background A , normalized slope B_1 , curvature B_2 , and support-diagnosed boundary enhancement H —not microscopic constants. DOI-grouped development reaches pooled $R^2 = 0.859300$ on 247 entities and 17,704 query rows, with 238/247 entities above 0.85, median/p95/maximum NRMSE 0.02958/0.21186/1.74495, and 748 negative predictions. The inverse-square-root atom recurs in 5/5 folds, but only 1–3 training entities per fold route to it; entity/DOI bootstrap lower bounds 0.82526/0.81641 remain below the endpoint.

The terminal 5-fold $\times$ 5-seed audit passes leakage checks and finds more reproducible function than raw-latent geometry (mean Spearman 0.9230 vs. 0.6056), yet both learned-bridge and GIRD gates fail. Decoder-functional spread R^2 is 0.4656; in four-support inference, conditional GIRD median NRMSE on the consensus prior-eligible subset is 0.0874, worse than direct- target dictionary 0.0497 and $\lambda = 0$ support OMP 0.0537 on that subset. Non-recurrent atoms, 0/300 passing sufficient-margin certificates, and a failed gauge audit locate the problem in sparse-prior identifiability (Appendix A). The direct expression still reaches $R^2 = 0.8593$, so this rejects the decoder prior rather than the response-coordinate interface.

Single-use Crystal-Cp result. The 2015–2019 DOI/InChIKey-disjoint cohort is reserved under a single-use contract. The formal 25-cell matrix and all-development packages are terminal; a target-blind seal binds 27 methods before any response is opened. One evaluation is authorized. GIRD remains diagnostic, and the frozen expression is unchanged even if it misses $R^2 = 0.85$.

6 LIMITATIONS AND CONCLUSION

Coordinates depend on probes, basis, weights, units, and support; they are not unique or causal. Misspecification and collinearity can hide or destabilize coefficients despite high curve R^2 . All domains need target-valued support, local methods can win, and Crystal’s prior fails; conditioning, stability, tails, negative predictions, and baselines therefore matter.

Within a frozen basis/probe specification, response coordinates are invariant to prediction-equivalent charts; support calibration is affine-equivariant under stated conditions. This is an inference protocol, not universal canonicalization. Despite Crystal’s failed prior, support-only ZT and vapor equations transfer at $R^2 = 0.9888$ and 0.9996 , with uncertainty and tails.

AI USE STATEMENT

Generative AI tools assisted with theoretical and conceptual development, mathematical claims and proof drafts, hypothesis and experiment design, method implementation, data cleaning, result interpretation, software and figure generation, literature search and synthesis, reference formatting, and drafting and editing the manuscript. AI-assisted numerical claims were checked against frozen raw artifacts and deterministic analysis code; generated code was subjected to focused tests, leakage checks, hash-bound manifests, and independent review where reported. The authors take responsibility for the final content, including all text, claims, code, and artifacts produced with AI assistance.

REPRODUCIBILITY STATEMENT

Current repository artifacts include frozen cohort and decision manifests, entity/DOI/time split definitions, support/query and target-perturbation audits where reported, repository-local source SHA-256 values, and deterministic analysis and figure generators. Dataset snapshots and local source files are identified by content hashes rather than mutable upstream links. The repository also contains the exact official-template build command. An indexed, allowlisted anonymous supplement was verified from its own root: every bundled hash passes and it rebuilds a byte-identical manuscript. It includes compact claim evidence but not third-party source archives or training checkpoints; their provenance and content hashes are supplied separately.

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A FROZEN DOMAIN PROTOCOLS

For Crystal multi-response OMP, step k retains margin Δ_k , residual R_k , and the largest aligned response displacement ϵ across seeds and support offsets. The preregistered conservative sufficient condition is $\Delta_k > 4\|R_k\|_F\epsilon + 2\epsilon^2$; failure is not a necessary instability result. Across 25 cells, median/minimum decoder-projection R^2 are 0.9967/0.0665, motifs recur in at most one or two of five folds per seed, 0/300 certificates pass, and the affine-gauge audit fails.

Table 4 makes explicit the coordinate conventions that are deliberately suppressed by a raw latent code. Temperatures are sorted before support selection. The headline ZT and vapor-pressure least-squares fits are unweighted float64 fits with `numpy.linalg.lstsq(rcond=None)`. The Crystal boundary fitter instead scales support columns and solves the full-rank system by explicit float64 SVD. The confirmed ZT and vapor-pressure support matrices have at least five distinct temperatures and numerical rank three for their three named columns.

Structure decisions use development query targets and are distinct from external support-only inference. ZT fixes the quadratic endpoint before its five entity folds; it is not selected against its base-lines. Vapor compares the coarse Clausius–Clapeyron basis with exactly one of $\{\log(T/T_r), (T - T_r)/T_r, (1/T - 1/T_r)^2\}$ in five DOI-grouped folds. Lowest median entity physical NRMSE wins; candidates within 1% use that displayed order. This selects $\log(T/T_r)$. The structure, reference-temperature rule, support mask, metric, tie break, and analysis are then frozen before temporal targets are opened; an external entity supplies only its support coefficients.

Table 3: Vapor development selection audit. Each T_r is the outer-training temperature median. Entries are held-out entity median physical NRMSE; the decision uses one median over all 282 entity values, not a median of these fold medians.

Fold	T_r (K)	coarse	$v_{\log}$	v_T	v_{1/T^2}
0	360.000	0.08137	0.01105	0.03073	0.03631
1	367.900	0.04556	0.01015	0.01223	0.01035
2	362.860	0.04135	0.01555	0.01802	0.01591
3	360.445	0.03867	0.00548	0.00871	0.00716
4	363.090	0.02487	0.01243	0.01263	0.01214
All entities	—	0.04349	0.01105	0.01420	0.01258

Table 4: Frozen real-domain inference details. “Every fourth” means zero-based ordered indices 0, 4, 8, . . . are support and all remaining rows are hidden queries. Decoder probes apply only to the neural functional ablations; the headline expressions are fitted directly from support.

Domain	Support rule	Named response coordinates	Decoder probes / seed aggregation
Starry ZT	Every fourth temperature; minimum five support rows.	$[1, \tau, \tau^2]$, where $\tau = (T - 280.584300 \text{ K}) / (193.386207 \text{ K})$; support-only quadratic fit.	41 equally spaced temperatures over the entity grid, equal weights; neural OOF uses each outer-training fold’s temperature mean and standard deviation, then a pointwise median over three seeds.
Vapor pressure	Every fourth temperature; minimum five support rows.	$[1, 1/T - 1/T_r, \log(T/T_r)]$ in $\log(P/1 \text{ kPa})$, with $T_r = 362.72 \text{ K}$; support-only fit.	41 equally spaced temperatures over the entity grid, equal weights; neural confirmation uses the frozen T_r and a pointwise median of three physical-pressure predictions; development stability instead uses outer-training-fold median T_r . The strongest local baseline is PCHIP applied to support log pressure.
Crystal C_p	Every fourth temperature (spread support); all other rows are queries.	Ordinary branch: five-term Shomate basis. Boundary branch: $[1, u, u^2, (\delta + 1 - u)^{-1/2}]$, with the nested degree-one subcase and $u = (T - T_{\min,e}) / (T_{\max,e} - T_{\min,e})$.	Headline development expression is deterministic. CNP and no-q MLP are pointwise medians over five seeds; the neural/GIRD registry remains unopened for confirmation.

For Crystal routing, let the ordered spread-support slopes be $s_i = (C_{p,i+1} - C_{p,i}) / (T_{i+1} - T_i)$ and

$$\rho_e = \frac{\max(0, s_{\text{last}})}{\max\{\text{median}_{i < \text{last}} |s_i|, 10^{-8} \max(1, \text{median} |C_p|) / (T_{\max,e} - T_{\min,e})\}}. \quad (10)$$

The boundary family is used when $\rho_e > \gamma$. The reported OOF estimate uses outer-training selections: fold 1 selects $(\gamma, d, \delta) = (200, 1, 0.003)$, folds 2–4 select $(100, 2, 0.0003)$, and fold 5 selects $(50, 2, 0.0003)$; every fold selects the inverse-square-root atom. The unopened final package separately freezes $(\gamma, d, \delta) = (100, 2, 0.0003)$.

For all domains, entity NRMSE is

$$\text{NRMSE}_e = \frac{\sqrt{|Q_e|^{-1} \sum_{i \in Q_e} (y_i - \hat{y}_i)^2}}{\sqrt{|Q_e|^{-1} \sum_{i \in Q_e} (y_i - \bar{y}_{Q_e})^2}}, \quad (11)$$

so the denominator is the population standard deviation (ddof=0). The frozen Starry evaluation manifest, ThermoML single-use lock and source-hash ledger, and Crystal single-use confirmation contract bind the corresponding support/query tables, evaluators, and source contents by SHA-256.